

EVALUATION-2 PREP

Phase 3 Study Guide

Topics Covered	Syllogism
Format	Theory → Statement Types → Rules → Worked Examples → Practice MCQs
Exam Pattern	25 Questions 45 Minutes
Color Theme	Green — Phase 3

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TOPIC: SYLLOGISM

1. What is Syllogism?

Syllogism is the logical process of deriving conclusions from given statements. You are given 2–3 **premises** (statements) and asked which of the provided conclusions **necessarily follow**. The key rule: *conclusions must be 100% true based on the premises — NOT just possible or probable.*

1.1 The Four Statement Types

Type	Code	Form	Venn Diagram Meaning	Example
Universal Positive	A	All S are P	Circle S is fully inside Circle P	All cats are animals
Universal Negative	E	No S is P	Circles S and P are completely separate	No cats are a dog
Particular Positive	I	Some S are P	Circles S and P partially overlap	Some cats are pets
Particular Negative	O	Some S are not P	Part of S is outside P	Some cats are not pets

1.2 Venn Diagram Approach — Step by Step

The most reliable method for syllogism. Draw circles for each term and fill in the relationships described by the premises. Then check each conclusion:

Step 1: Identify all terms (subjects and predicates) in the statements.

Step 2: Draw the possible Venn diagram(s) — some statements allow multiple diagrams.

Step 3: For each conclusion, check if it is TRUE in ALL possible diagrams.

→ If TRUE in ALL diagrams: Conclusion follows (mark ✓)

→ If TRUE in SOME diagrams only: Does NOT definitely follow (mark ✗)

→ If FALSE in ALL diagrams: Definitely does not follow (mark ✗)

1.3 Conversion Table — Immediate Inferences

Sometimes you need to convert a statement to check conclusions. Conversions that are always valid:

Original Statement	Valid Conversion	Type Change
All S are P (A)	Some P are S (I)	A → I (by conversion, particular)
No S is P (E)	No P is S (E)	E → E (fully reversible)
Some S are P (I)	Some P are S (I)	I → I (reversible)
Some S are not P (O)	Cannot be converted validly	O has NO valid converse

1.4 Syllogism Rules — Conclusion from Two Premises

When combining two premises with a common middle term (M), use these rules:

Premise 1 (P1)	Premise 2 (P2)	Valid Conclusion	Notes
A (All S are M)	A (All M are P)	A: All S are P	Two universals → Universal conclusion
A (All S are M)	E (No M is P)	E: No S is P	Universal + Universal Neg → Negative
A (All S are M)	I (Some M are P)	I: Some S are P	Universal + Particular → Particular
A (All S are M)	O (Some M are not P)	O: Some S are not P	—

E (No S is M)	A (All M are P)	O: Some P are not S	via conversion
E (No S is M)	I (Some M are P)	O: Some P are not S	—
I (Some S are M)	A (All M are P)	I: Some S are P	Particular premise → Particular conclusion
I (Some S are M)	E (No M is P)	O: Some S are not P	—
O + anything	—	No definite conclusion	O premise never gives definite result
I + I	—	No definite conclusion	Two particulars → no conclusion

Key Thumb Rules:

- **Two negative premises** → No conclusion.
- **Two particular premises** → No conclusion.
- **One negative premise** → Conclusion must be negative.
- **One particular premise** → Conclusion must be particular.
- **Two universal premises** → Conclusion can be universal.
- If both premises are positive → Conclusion is positive.

1.5 Complementary Pairs (Either-Or Conclusions)

When neither conclusion follows individually but together they form a complementary pair, the answer is 'Either I or II follows'. A complementary pair occurs when:

- One conclusion is **I-type** (Some A are B) and the other is **O-type** (Some A are not B) — for same A and B
- One is **A-type** (All A are B) and the other is **O-type** (Some A are not B) — for same A and B
- One is **I-type** and the other is **E-type** (No A is B) — for same A and B

■■ **Important:** Both conclusions must deal with the SAME pair of terms to form a complementary pair.

1.6 Worked Examples — Syllogism

(From HitBullseye Syllogism PDF)

Q1.

Statements:

1. All dogs are animals.
2. All animals are living beings.

Conclusions:

- I. All dogs are living beings.
- II. Some living beings are dogs.

Solution: Statement 1: A (All dogs $\rightarrow$ animals). Statement 2: A (All animals $\rightarrow$ living beings).

A + A $\rightarrow$ A: All dogs are living beings. ✓ Conclusion I follows.

Convert conclusion I: Some living beings are dogs (I-type). ✓ Conclusion II also follows.

Both Conclusions I and II follow.

Q2.

Statements:

1. All roses are flowers.
2. No flower is red.

Conclusions:

- I. No rose is red.
- II. Some roses are red.

Solution: Statement 1: A (All roses $\rightarrow$ flowers). Statement 2: E (No flower $\rightarrow$ red).

A + E $\rightarrow$ E: No rose is red. ✓ Conclusion I follows.

Conclusion II (Some roses are red) contradicts I. ✗

Only Conclusion I follows.

Q3.

Statements:

1. Some pens are books.
2. All books are pencils.

Conclusions:

- I. Some pens are pencils.
- II. Some pencils are pens.
- III. All pencils are pens.

Solution: Stmt 1: I (Some pens $\rightarrow$ books). Stmt 2: A (All books $\rightarrow$ pencils).

I + A $\rightarrow$ I: Some pens are pencils. ✓ Conclusion I follows.

Convert I: Some pencils are pens. ✓ Conclusion II follows.

Conclusion III (All pencils are pens): Not derivable — only 'some' is known. ✗

Conclusions I and II follow.

Q4.

Statements:

1. No table is a chair.

2. Some chairs are desks.

Conclusions:

I. No table is a desk.

II. Some desks are not tables.

Solution: Stmt 1: E (No table $\rightarrow$ chair). Stmt 2: I (Some chairs $\rightarrow$ desks).

E + I $\rightarrow$ O: Some desks are not tables (by conversion: some chairs=desks, no chair=table $\rightarrow$ some desks not table). ✓

Conclusion I: No table is a desk — this would be E-type, stronger claim not supported. ✗

Only Conclusion II follows.

Q5.

Statements:

1. All cats are dogs.

2. Some dogs are birds.

Conclusions:

I. Some cats are birds.

II. Some birds are cats.

Solution: Stmt 1: A (All cats $\rightarrow$ dogs). Stmt 2: I (Some dogs $\rightarrow$ birds).

The 'some dogs' that are birds may or may not include cats (the Venn diagram allows both cases).

Conclusion I: Not definite — cats are dogs but the 'some dogs that are birds' need not include cats. ✗

Conclusion II: Similarly not definite. ✗

I and II form a possible complementary pair (I and O-type for same terms)? No — check: I = 'some cats are birds', II = 'some birds are cats'. These are converses of each other, not complementary.

Neither conclusion definitively follows.

Neither Conclusion follows.

Q6.

Statements:

1. Some students are intelligent.

2. All intelligent persons are hardworking.

Conclusions:

I. Some students are hardworking.

II. All hardworking persons are students.

Solution: Stmt 1: I (Some students $\rightarrow$ intelligent). Stmt 2: A (All intelligent $\rightarrow$ hardworking).

I + A $\rightarrow$ I: Some students are hardworking. ✓ Conclusion I follows.

Conclusion II (All hardworking $\rightarrow$ students): Way stronger, not derivable. ✗

Only Conclusion I follows.

Q7.

Statements:

1. No bird is a fish.

2. No fish is a mammal.

Conclusions:

I. No bird is a mammal.

II. Some mammals are birds.

Solution: Stmt 1: E. Stmt 2: E. Two negative premises → No definite conclusion.

Conclusion I: Not derivable (birds and mammals could overlap or not — unknown). ✗

Conclusion II: Not derivable. ✗

I and II form a complementary pair? I = E-type, II = I-type for bird-mammal. Yes — complementary pair!

Either Conclusion I or II follows (Complementary pair).

Q8.

Statements:

1. All mangoes are fruits.
2. All fruits are sweet.
3. Some sweets are tasty.

Conclusions:

- I. All mangoes are sweet.
- II. Some tasty things are fruits.
- III. Some sweets are mangoes.

Solution: Stmts 1+2: A+A → All mangoes are sweet. ✓ Conclusion I follows.

Stmts 2+3: A + I (backward) → Some sweets are fruits (by reversing stmt3 not valid this way). Stmt3: Some sweets are tasty — 'some sweets' are tasty but we don't know if those are fruits. ✗ Conclusion II does not follow.

Stmt 1+2: All mangoes are sweet. Convert: Some sweets are mangoes. ✓ Conclusion III follows.

Conclusions I and III follow.

PRACTICE MCQs — Phase 3: Syllogism (with Solutions)

15 syllogism questions. Target: 13 minutes (55 sec/question). Draw Venn diagrams quickly for each.

1. Statements: All pens are books. All books are bags.

Conclusions: I. All pens are bags. II. Some bags are pens.

- A) Only I follows
- B) Only II follows
- C) Both I and II follow
- D) Neither follows
- ✓ C) Both I and II follow

Explanation: $A+A \rightarrow A$: All pens are bags (I ✓). Convert: Some bags are pens (II ✓).

2. Statements: Some birds are flowers. No flower is red.

Conclusions: I. No bird is red. II. Some birds are not red.

- A) Only I
- B) Only II
- C) Both
- D) Neither
- ✓ B) Only II

Explanation: $I + E \rightarrow O$: Some birds are not red (II ✓). Conclusion I claims ALL birds not red — too strong, not derivable. ✗

3. Statements: All doctors are engineers. Some engineers are teachers.

Conclusions: I. Some doctors are teachers. II. Some teachers are doctors.

- A) Both follow
- B) Only I
- C) Only II
- D) Neither
- ✓ D) Neither

Explanation: $A + I \rightarrow I$: Some doctors are teachers? Only if those 'some engineers who are teachers' include doctors — not certain. Neither conclusion definitely follows.

4. Statements: No table is wood. All wood is brown.

Conclusions: I. No table is brown. II. Some brown things are not tables.

- A) Only I
- B) Only II
- C) Both
- D) Neither
- ✓ B) Only II

Explanation: $E + A \rightarrow O$: Some brown things are not tables (II ✓). Conclusion I (No table is brown) would require $E+A \rightarrow E$ but that's not a valid syllogism rule here. Only II follows.

5. Statements: All cats are dogs. No dog is a lion.

Conclusions: I. No cat is a lion. II. Some lions are cats.

- A) Only I
- B) Only II
- C) Both
- D) Either I or II

✓ A) Only I

Explanation: $A + E \rightarrow E$: No cat is a lion (I ✓). Conclusion II contradicts I. Only I follows.

6. Statements: Some houses are buildings. All buildings are structures.

Conclusions: I. All houses are structures. II. Some houses are structures.

- A) Only I
- B) Only II
- C) Both
- D) Neither

✓ B) Only II

Explanation: $I + A \rightarrow I$: Some houses are structures (II ✓). Conclusion I (All houses $\rightarrow$ structures) is too strong. Only II follows.

7. Statements: No man is perfect. Some humans are men.

Conclusions: I. Some humans are not perfect. II. No human is perfect.

- A) Only I
- B) Only II
- C) Both
- D) Either I or II

✓ A) Only I

Explanation: $E(\text{No man perfect}) + I(\text{Some humans are men}) \rightarrow O$: Some humans are not perfect (I ✓). II is too strong. Only I.

8. Statements: All A are B. All B are C. Some C are D.

Conclusions: I. Some A are D. II. All A are C.

- A) Both
- B) Only II
- C) Only I
- D) Neither

✓ B) Only II

Explanation: $A+A \rightarrow$ All A are C (II ✓). Some C are D — the 'some C' may or may not include A. I not definite. Only II.

9. Statements: No fruit is vegetable. Some vegetables are tasty.

Conclusions: I. Some tasty things are not fruits. II. No fruit is tasty.

- A) Only I
- B) Only II
- C) Both
- D) Neither

✓ A) Only I

Explanation: $E + I \rightarrow O$: Some tasty things are not fruits (I ✓). Conclusion II (No fruit is tasty) — cannot conclude from given info. Only I.

10. Statements: Some books are pens. Some pens are erasers.

Conclusions: I. Some books are erasers. II. Some erasers are books.

- A) Both
- B) Only I
- C) Only II
- D) Neither
- ✓ D) Neither

Explanation: I + I → No definite conclusion. Both I and II are uncertain. Neither follows.

11. Statements: All cars are vehicles. Some vehicles are trucks.

Conclusions: I. Some trucks are cars. II. Some cars are trucks.

- A) Both
- B) Only I
- C) Only II
- D) Neither
- ✓ D) Neither

Explanation: A + I (reversed middle): The some vehicles that are trucks may not be cars. Neither I nor II definitely follows.

12. Statements: All politicians are corrupt. No corrupt person is respected.

Conclusions: I. No politician is respected. II. Some respected persons are not politicians.

- A) Only I
- B) Only II
- C) Both
- D) Either I or II
- ✓ C) Both

Explanation: A + E → E: No politician is respected (I ✓). Convert E: No respected person is a politician → Some respected not politicians (II ✓). Both follow.

13. Statements: Some students pass. All who pass are promoted.

Conclusions: I. Some students are promoted. II. All students are promoted.

- A) Only I
- B) Only II
- C) Both
- D) Neither
- ✓ A) Only I

Explanation: I + A → I: Some students are promoted (I ✓). II is too strong (only 'some' students pass). Only I.

14. Statements: No star is a planet. All planets are celestial bodies.

Conclusions: I. No star is a celestial body. II. Some celestial bodies are not stars.

- A) Only I
- B) Only II
- C) Both
- D) Neither

✓ B) Only II

Explanation: $E + A \rightarrow O$: Some celestial bodies are not stars (II ✓). I says no star is celestial — too strong, cannot conclude. Only II.

15. Statements: All kings are queens. Some queens are jokers.

Conclusions: I. Some kings are jokers. II. Some jokers are kings.

- A) Both
- B) Only I
- C) Only II
- D) Neither

✓ D) Neither

Explanation: $A + I \rightarrow I$? The 'some queens who are jokers' may or may not include kings. Not definite. Neither follows. Complementary pair check: I and II are converses, not complementary. Neither follows.